WORKED SOLUTION

STEP I 1987, Question 2

Pure • Areas of Circle Packings

STEP I 1987, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q2

The key idea

Join the centres of the small circles. They all lie on the circle of radius R, midway between the two big circles, and they form a regular n-gon whose sides pass through the points where neighbouring small circles touch. That polygon is the dividing line between the pieces called Y (inside it) and X (outside it). So every area comes from three simple shapes: the polygon, a disc, and sectors of the small circles cut off by the polygon's corners.

Solution

The relationship between n, R and r

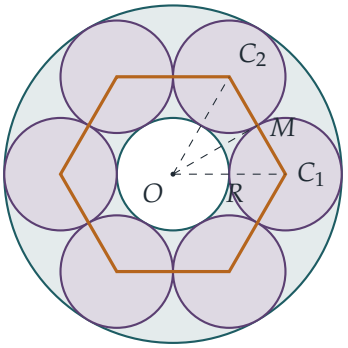
Each small circle touches both big circles, so its centre is at distance $R - r + r = R$ from the common centre O: all n centres lie on the circle of radius R. By symmetry they are equally spaced, so neighbouring centres C_1, C_2 subtend an angle $2\pi/n$ at O.

Two neighbouring circles touch, so $C_1C_2 = 2r$. In the isosceles triangle OC_1C_2, with $OC_1 = OC_2 = R$, dropping the perpendicular from O to the midpoint M of C_1C_2 gives

$$r = C_1M = R\sin\frac{\pi}{n}.$$

This is the required relationship. (We need $n \geq 3$; for $n = 2$ the inner circle would have radius $R - r = 0$.) The same triangle gives

$$OM = R\cos\frac{\pi}{n} = \sqrt{R^2 - r^2}.$$



The figure (drawn for $n = 6$) shows the polygon of centres in amber. Its sides pass through the points of contact of neighbouring circles, such as M.

The area Y

The polygon is made of n triangles congruent to OC_1C_2, each of area $\frac{1}{2} \cdot 2r \cdot \sqrt{R^2 - r^2}$. So

$$\text{area of polygon} = nr\sqrt{R^2 - r^2}.$$

What lies inside the polygon? Its inradius is $OM = \sqrt{R^2 - r^2}$, and $R^2 - r^2 \geq (R - r)^2$ because this rearranges to $2r(R - r) \geq 0$. So the inner disc of radius $R - r$ lies inside the polygon. At each vertex there is a sector of a small circle. The interior angle of a regular n-gon is $\pi - \frac{2\pi}{n}$, so each such sector has area

$$\tfrac{1}{2}r^2\left(\pi - \frac{2\pi}{n}\right) = \pi r^2\left(\frac{1}{2} - \frac{1}{n}\right).$$

These n sectors and the inner disc don't overlap one another, since the small circles only touch each other and the inner circle. What is left of the polygon is exactly the n curved triangles next to the inner circle, and together these make up Y:

$$Y = nr\sqrt{R^2 - r^2} - \pi(R - r)^2 - n\pi r^2\left(\frac{1}{2} - \frac{1}{n}\right),$$

as required.

Ask yourself Why is the side C_1C_2 the right place to split X from Y?

The side C_1C_2 runs from centre to centre through the point of contact M, so it never crosses the gaps between the circles. Every gap next to the inner circle lies on the O side of it, and every gap next to the outer circle lies on the far side. Cutting along the polygon therefore separates the two kinds of gap cleanly, and it cuts each small circle along two radii, which is why we get sectors.

The areas X and Z

Outside the polygon but inside the outer circle, we remove the rest of each small circle. That rest is the reflex sector of angle $2\pi - (\pi - \frac{2\pi}{n}) = \pi + \frac{2\pi}{n}$, of area $\pi r^2\left(\frac{1}{2} + \frac{1}{n}\right)$.

Hence

$$X = \pi(R+r)^2 - nr\sqrt{R^2-r^2} - n\pi r^2\left(\frac{1}{2}+\frac{1}{n}\right).$$

The n discs of radius r give

$$Z = n\pi r^2.$$

The limit as $n \to \infty$

Adding, the polygon terms cancel and the sector terms combine to $n\pi r^2$:

$$\begin{aligned} X+Y &= \pi(R+r)^2 - \pi(R-r)^2 - n\pi r^2 \\ &= 4\pi Rr - n\pi r^2. \end{aligned}$$

This is just the area of the annulus minus the area of the circles, as it should be. Therefore

$$\frac{X+Y}{Z} = \frac{4\pi Rr - n\pi r^2}{n\pi r^2} = \frac{4R}{nr} - 1 = \frac{4}{n\sin(\pi/n)} - 1,$$

using $r = R\sin(\pi/n)$. As $n \to \infty$, $\pi/n \to 0$ and $\dfrac{\sin(\pi/n)}{\pi/n} \to 1$, so $n\sin(\pi/n) \to \pi$ and

$$\frac{X+Y}{Z} \to \frac{4}{\pi} - 1 \approx 0.273.$$

Checks

- For $n = 3$, $R = 1$ we get $r = \frac{\sqrt{3}}{2}$, $Y \approx 0.0646$ and $X \approx 3.7497$. A numerical integration over a fine polar grid, classifying each point as inside a circle, inside the polygon or outside it, agreed to four significant figures for $n = 3, 5, 8, 12$.
- The ratio $\frac{4}{n\sin(\pi/n)} - 1$ is 0.2944 at $n = 10$, 0.27345 at $n = 100$ and 0.273240 at $n = 1000$. It is closing in on $4/\pi - 1 = 0.273240\ldots$.
- Heuristically, for large n the circles look like a row of discs of diameter $2r$ in a strip of width $2r$. A disc fills πr^2 of each $2r \times 2r$ square, and the gaps fill $4r^2 - \pi r^2$, giving the ratio $(4-\pi)/\pi$.

Common slips

- Using the exterior angle $2\pi/n$ of the polygon for the sectors instead of the interior angle $\pi - 2\pi/n$.
- Forgetting that the inner disc must be removed in Y, or not checking that it really lies inside the polygon.

Beyond STEP

The final limit is $\lim_{x\to 0}\frac{\sin x}{x} = 1$ in disguise. It is the same fact that lets Archimedes squeeze π between perimeters of inscribed and circumscribed polygons. The question is also a toy version of a circle-packing density problem, here asking what fraction of a thin annulus the circles can fill. To make limits like this rigorous, try Abbott, *Understanding Analysis*.

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